COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

STAT 5703

Statistical Inference and Modeling for Data Science

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Confidence Intervals
aka Interval Estimators

Definition

A *confidence interval* for a parameter θ is a pair of statistics $[l(X_1, \ldots, X_n), u(X_1, \ldots, X_n)]$ such that

$$P(l(X_1, \ldots, X_n) < \theta < u(X_1, \ldots, X_n)) = 1 - \alpha$$

where $1 - \alpha$ is said to be the confidence level of the interval.

More rigorous definition:

A $(1 - \alpha)\%$ *confidence set* $C = C(X_1, \ldots, X_n)$ is such that

$$P(\theta \in C) \geq 1 - \alpha, \forall\, \theta$$

Note: θ is fixed, and C is random!

Generic Form

Very often (but not always), the confidence intervals have the following generic form:

l = Estimator – (Critical Value)*SE

u = Estimator + (Critical Value)*SE

or

$$\boxed{\hat{\theta} \pm (\text{Crit. Value}) * \text{SE}(\hat{\theta}) = \hat{\theta} \pm ME}$$

Note: The part $(\text{Crit. Value}) * \text{SE}(\hat{\theta})$ is called *margin of error* (ME) of the CI

Q: How are these formulas obtained?

Example 1: CI for μ when σ is known

Let

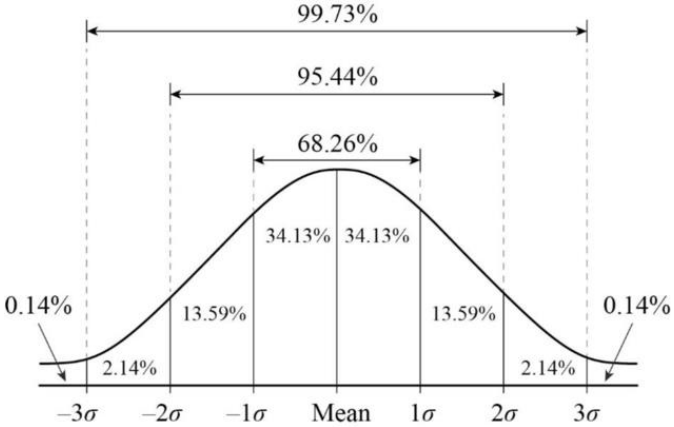
$$X_1, \ldots, X_n \sim \mathrm{N}(\mu, \sigma)$$

Recall:

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$$

Suppose $\alpha = 0.05$, so we want a 95% CI.

Then the critical value is $z_{\frac{\alpha}{2}} = 1.96$



Example 1 (solution):

$$\Rightarrow \quad P\left(-1.96 < Z < 1.96\right) = 0.95$$

solve for μ

$$\Leftrightarrow \quad P\left(-1.96 < \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} < 1.96\right) = 0.95$$

$$\Leftrightarrow \quad P\left(-1.96\frac{\sigma}{\sqrt{n}} < \bar{x} - \mu < 1.96\frac{\sigma}{\sqrt{n}}\right) = 0.95$$

Example 1 (Solution continued)

From previous slide:

$$\mathrm{P}\left(-1.96\frac{\sigma}{\sqrt{n}} < \bar{x} - \mu < 1.96\frac{\sigma}{\sqrt{n}}\right) = 0.95$$

subtract $\bar{x}$ from each side of the inequality:

$$\Leftrightarrow \mathrm{P}\left(-1.96\frac{\sigma}{\sqrt{n}} - \bar{x} < -\mu < 1.96\frac{\sigma}{\sqrt{n}} - \bar{x}\right) = 0.95$$

Example 1 (Solution continued)

$$\Leftrightarrow P\left(-1.96\frac{\sigma}{\sqrt{n}} - \bar{x} < -\mu < 1.96\frac{\sigma}{\sqrt{n}} - \bar{x}\right) = 0.95$$

$$\Leftrightarrow P\Big(\underbrace{\bar{x} - 1.96\frac{\sigma}{\sqrt{n}}}_{l(x)} < \mu < \underbrace{\bar{x} + 1.96\frac{\sigma}{\sqrt{n}}}_{u(x)}\Big) = .95$$

95% CI for μ is $\bar{x} \pm 1.96\frac{\sigma}{\sqrt{n}} = SE(\bar{x})$

estimator crit. value

Example 1 (Solution continued):

CI for μ when σ is known:

$$\boxed{\bar{x} \pm z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}}$$

Suppose now you got normally distributed data that resulted in $\bar{x} = 80, n = 31$ and a known population standard deviation $\sigma = 2$. Compute the 95% CI for the population mean μ.

$\Rightarrow$ 95% CI for μ is

$$80 \pm 1.96 \frac{2}{\sqrt{31}} = 80 \pm 0.7 = \boxed{(79.3, 80.7)}$$

Exercise 1:

Suppose that $X_1, \ldots, X_n \sim \mathrm{N}(\mu, \sigma)$, where σ is known. What is the confidence level of the following CI for μ:

$$\left(\bar{X} - 0.88\frac{\sigma}{\sqrt{n}}, \bar{X} + 1.91\frac{\sigma}{\sqrt{n}}\right)$$

Answer: 78%

Interpretation

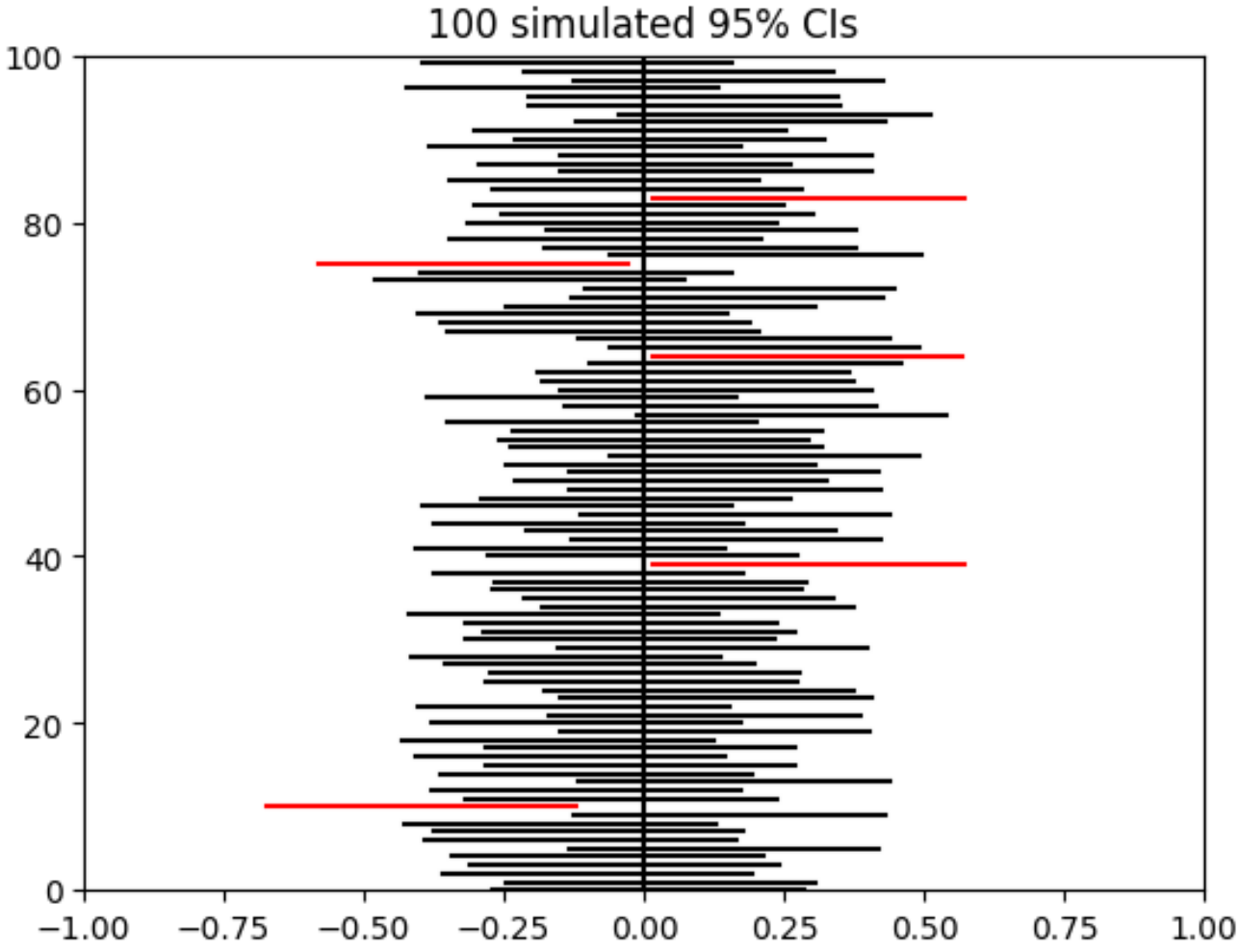
Q: What does 95% confidence mean?

A: The procedure we are using to construct the interval is 95% reliable. That is, 95% of all the intervals constructed with this method will cover the unknown value of μ, and 5% will miss it.

Therefore, we don't say $\mu \in (l, u)$ with *probability* 95%, because μ is not a random variable! The best way express the interpretation is to say:

We are 95% *confident* that μ is between l and u.

Interpretation



Note on the width of the CI

The width of the confidence interval, that is, its accuracy depends on:

- The sample size *n*: the larger the sample size the narrow the width of the CI.
- The confidence level: the higher the confidence the wider the CI will be!
- The standard deviation of the population or SE: the larger the SE the wider the CI will be.
- The method used to construct the CI

Exercise 2

Consider a random sample $X_1, \ldots, X_n \sim \mathrm{N}(\mu, \sigma)$, where the population variance σ^2 is known. Show that

$$\left(-\infty, \bar{X} - 0.126\frac{\sigma}{\sqrt{n}}\right) \cup \left(\bar{X} + 0.126\frac{\sigma}{\sqrt{n}}, \infty\right)$$

is a confidence set for the unknown mean μ with a confidence level 90%.

General Method for Constructing CIs

- We need to find a function $h(x_1, \ldots, x_n, \theta)$ which depends on *both* the sample *and* the parameter, such that the distribution of h does *not* depend on θ. Such a function is known as *pivotal quantity* or *pivot*.

- Once we have the pivotal quantity, we need to solve for θ the inequalities:

$$\mathrm{CV}_{1-\frac{\alpha}{2}} < h(x_1, \ldots, x_n, \theta) < \mathrm{CV}_{\frac{\alpha}{2}}$$

- Note that CV_α means the quantile of the distribution of h, corresponding to *upper* area α!

Example 2

Let

$$X_1, \ldots, X_n \sim \text{Exp}(\lambda)$$

Then it is known that $\lambda X_i \sim \text{Exp}(1)$ for all $i = 1, \ldots, n$

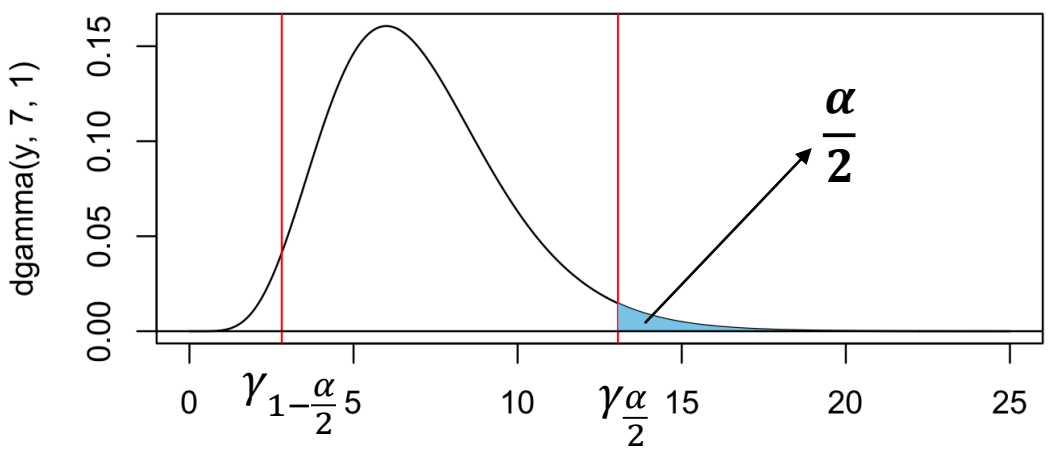
$$\Rightarrow h(X_1, \ldots, X_n, \lambda) = \underbrace{\lambda \sum_{i=1}^{n} X_i}_{\textbf{\textit{Pivot}}} \quad \sim \quad \underbrace{\Gamma(n, 1)}_{\textbf{\textit{Distribution does not depend on } \lambda}}$$

(Where does the above come from?)

Use $h(X_1, \ldots, X_n, \lambda) = \lambda \sum_{i=1}^{n} X_i$ as a pivot and derive a 95% CI for λ.

Example 2: Solution

1) Find the Gamma distribution quantiles $\gamma_{1-\frac{\alpha}{2}}$ and $\gamma_{\frac{\alpha}{2}}$ by either using a table or software.
Note the notation meaning of the subscript!



2) Solve $\mathrm{P}\left(\gamma_{1-\frac{\alpha}{2}} < \lambda \sum_{i=1}^{n} X_i < \gamma_{\frac{\alpha}{2}}\right) = 1 - \alpha$ for λ

Example 2: Solution

$$\Longleftrightarrow \quad P\left(\underbrace{\frac{\chi_{1-\frac{\alpha}{2}}}{\sum x_i}}_{l} < \lambda < \underbrace{\frac{\chi_{\frac{\alpha}{2}}}{\sum x_i}}_{u} \right) = 1-\alpha$$

CI for λ is $\boxed{\left(\frac{\chi_{1-\frac{\alpha}{2}}}{\sum x_i}, \frac{\chi_{\frac{\alpha}{2}}}{\sum x_i} \right)}$

Exercise 3

Let

$$X_1, \ldots, X_n \sim \mathrm{U}(0, \theta)$$

Define $Y = \max\{X_1, \ldots, X_n\}$.

Then $h(X_1, \ldots, X_n, \theta) = \frac{Y}{\theta}$ is a pivot with pdf

$$f(h) = nh^{n-1}, \text{ for } 0 \leq h \leq 1$$

Reason: $\mathrm{P}\left(\frac{Y}{\theta} < h\right) = \mathrm{P}(Y < \theta h) = \frac{(\theta h)^n}{\theta^n} = h^n$

Use the above pivot to construct a CI.

Wald CI

Recall that under regularity conditions, as $n \to \infty$,

$$\frac{\hat{\theta} - \theta}{\widehat{\mathrm{SE}}(\hat{\theta})} \xrightarrow{d} N(0,1)$$

$$\Rightarrow P\left(-z_{\frac{\alpha}{2}} < \frac{\hat{\theta} - \theta}{\widehat{\mathrm{SE}}(\hat{\theta})} < z_{\frac{\alpha}{2}}\right) \approx 1 - \alpha$$

Therefore, the Wald CI for a MLE of θ is:

$$\hat{\theta} \pm z_{\frac{\alpha}{2}} \widehat{\mathrm{SE}}(\hat{\theta})$$

Note: The $\widehat{\mathrm{SE}}(\hat{\theta})$ is usually obtained from the Fisher information.

Example 3

Let $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$ and $X = \sum_{i=1}^{n} X_i$. Then

$$X \sim \text{Binom}(n, p)$$

Derive the Wald CI for p

Solution:

MLE of p is $\hat{p} = \frac{X}{n}$

$$SE(\hat{p}) = \sqrt{Var(\hat{p})} = \sqrt{\frac{p(1-p)}{n}}$$

$$\Rightarrow \widehat{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Example 3 (Solution):

Therefore, the Wald confidence interval is:

$$\boxed{\hat{p} \pm z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}$$

Large sample assumption check rule of thumb:

$$n\hat{p} \geq 10 \;\&\; n(1-\hat{p}) \geq 10$$

This means, there should be at least 10 expected successes and 10 expected failures in a sample to use the normal distribution as an approximation.

Score CI for p:

Instead of estimating $\mathrm{SE}(\hat{p})$ use the exact SE:

$$\frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} \approx N(0,1)$$

$$\Rightarrow P\left(-z_{\frac{\alpha}{2}} < \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} < z_{\frac{\alpha}{2}}\right) \approx 1 - \alpha$$

Solve the quadratic inequality:

$$\hat{p} - p < z_{\frac{\alpha}{2}}\sqrt{\frac{p(1-p)}{n}}$$

$$\Rightarrow (\hat{p} - p)^2 < z_{\frac{\alpha}{2}}^2 \frac{p(1-p)}{n}$$

Score CI for *p*:

Solve the quadratic inequality for p to obtain:

$$\hat{p}^2 - 2p\hat{p} + p^2 < p\frac{z_{\frac{\alpha}{2}}^2}{n} - p^2\frac{z_{\frac{\alpha}{2}}^2}{n}$$

$$\Leftrightarrow \left(1 + \frac{z_{\frac{\alpha}{2}}^2}{n}\right)p^2 - \left(2\hat{p} + \frac{z_{\frac{\alpha}{2}}^2}{n}\right)p + \hat{p}^2 < 0$$

...

$$\frac{\hat{p} + \frac{z_{\frac{\alpha}{2}}^2}{2n}}{1 + \frac{z_{\frac{\alpha}{2}}^2}{n}} \pm z_{\frac{\alpha}{2}} \frac{\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z_{\frac{\alpha}{2}}^2}{4n^2}}}{1 + \frac{z_{\frac{\alpha}{2}}^2}{n}}$$

Note: For the most common 95% CI, $z_{\frac{\alpha}{2}}^2 \approx 1.92$ and $\frac{\hat{p}+\frac{z_{\frac{\alpha}{2}}^2}{2n}}{1+\frac{z_{\frac{\alpha}{2}}^2}{n}} \approx \frac{x+2}{n+4}$, which is the same as the Wilson correction to the $\hat{p}$

Exercise 3

According to a Field Poll, 400 out of 506 feel that "education and our schools" is one of the top issues facing California. We wish to construct a 90% confidence interval for the true proportion of California adults who feel that education and the schools is one of the top issues facing California. Apply both versions.

Example 4: CI for Gamma distribution

Let $X_1, \ldots, X_n$ be an iid sample from the $\Gamma(\alpha, \lambda)$ distribution. Recall that the MLE must be found numerically by solving

$$n\big(\log(\hat{\alpha}) - \log \bar{x} - digamma(\hat{\alpha})\big) + \sum_{i=1}^{n} \log(x_i) = 0$$

and then substituting into

$$\hat{\lambda} = \frac{\hat{\alpha}}{\bar{x}}$$

The Fisher information matrix is:

$$I(\alpha, \lambda) = n \begin{bmatrix} \psi_1(\alpha) & -\frac{1}{\lambda} \\ -\frac{1}{\lambda} & \frac{\alpha}{\lambda^2} \end{bmatrix}$$

Example 4: CI for Gamma distribution

Suppose a sample of size $n = 50$ resulted in:

$$\bar{x} = 2.4, \sum_{i=1}^{n} \log(x_i) = 40.1$$

Then solving numerically we obtain $\hat{\alpha} = 6.9, \hat{\lambda} = 2.9$.

The estimated Fisher information matrix is:

$$\hat{I} = 50 \begin{bmatrix} \psi_1(\hat{\alpha}) & -\frac{1}{\hat{\lambda}} \\ -\frac{1}{\hat{\lambda}} & \frac{\hat{\alpha}}{\hat{\lambda}^2} \end{bmatrix} \approx 50 \begin{bmatrix} 0.15 & -0.35 \\ -0.35 & 0.83 \end{bmatrix}$$

$$= \begin{bmatrix} 7.75 & -17.29 \\ -17.29 & 41.5 \end{bmatrix} \Rightarrow \hat{I}^{-1} = \begin{bmatrix} 1.83 & 0.76 \\ 0.76 & 0.34 \end{bmatrix}$$

Example 4: CI for Gamma distribution

Suppose a sample of size n = 50 resulted in

$$\hat{\alpha} = 6.9, \hat{\lambda} = 2.9$$
$$\hat{I}^{-1} = \begin{bmatrix} 1.83 & 0.76 \\ 0.76 & 0.34 \end{bmatrix}$$

Then the 95% Wald CI for α and λ are respectively:

$$6.9 \pm 1.96\sqrt{1.83} = (4.25, 9.55)$$
$$2.9 \pm 1.96\sqrt{0.34} = (1.76, 4.04)$$

Example 4: CI for Gamma distribution

Suppose a sample of size $n = 50$ resulted in

$$\hat{\alpha} = 6.9, \hat{\lambda} = 2.9$$

$$\hat{I}^{-1} = \begin{bmatrix} 1.83 & 0.76 \\ 0.76 & 0.34 \end{bmatrix}, \hat{I} = \begin{bmatrix} 7.75 & -17.29 \\ -17.29 & 41.5 \end{bmatrix}$$

Then the 95% Wald CI for α and λ are:

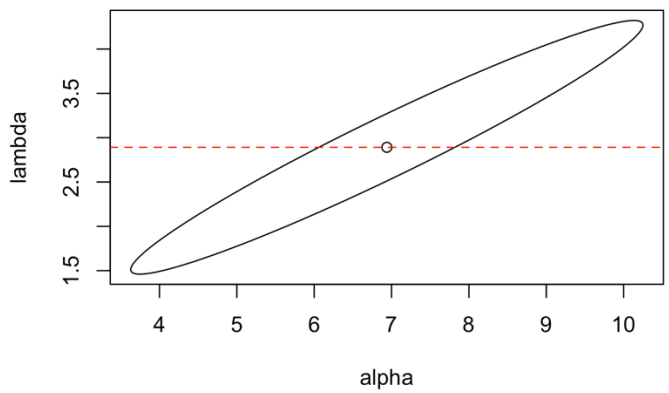
$$6.9 \pm 1.96\sqrt{1.83} = (4.25, 9.55)$$

$$2.9 \pm 1.96\sqrt{0.34} = (1.76, 4.04)$$

Note: The intervals are much narrower if we use reciprocal of Fisher:

$$6.9 \pm 1.96\sqrt{\frac{1}{7.75}} = (6.2, 7.6)$$

What about the off-diagonal element 0.76?
It can be used to construct a joint confidence ellipse.



The second interval does not account for uncertainty in the other parameter. It is computed for the fixed Value of the other parameter equal to its MLE

Example 5: CI for μ when σ is unknown

Recall that if $X_1, \ldots, X_n \sim \mathrm{N}(\mu, \sigma)$, then

$$h(x_1, \ldots, x_n, \mu) = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} \sim t_{n-1}$$

$$\Rightarrow P\left(-t_{\frac{\alpha}{2}} < \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} < t_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

Therefore, the CI for μ is

$$\bar{x} \pm t_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$$

See R code for a real data example.

Two - sample CI for difference of means

Let $X_1, \ldots, X_n \sim N(\mu_1, \sigma)$ and independently $Y_1, \ldots, Y_m \sim N(\mu_2, \sigma)$.

Note the two normal populations have the same unknown standard deviation!

Then a CI for $\mu_1 - \mu_2$ is

$$\bar{x} - \bar{y} \pm t_{\frac{\alpha}{2}} \times s_p \sqrt{\frac{1}{n} + \frac{1}{m}}$$

$$s_p = \sqrt{\frac{(n-1)s_x^2 + (m-1)s_y^2}{n+m-2}}$$

where df = $n + m - 2$

Profile Confidence Intervals

Choose the following set as CI:

$$S(\theta) = \left\{\theta \in \Theta\colon\ 2l_n\left(\hat{\theta}\right) - 2l_n(\theta) < \chi_\alpha^2\right\}$$

This is based on asymptotic theory of the likelihood ratio test statistics, covered in two lectures.

Example: Profile CI

Let $X_1, \ldots, X_n \sim$ Bernoulli(p) and $X = \sum_{i=1}^{n} X_i$. Then

$$X \sim \text{Binom}(n, p)$$

Derive the profile CI for p

Start with the likelihood:

$$L_n(p) = p^x (1-p)^{n-x}$$

Now obtain the log-likelihood at p and at $\hat{p}$:

$$l_n(p) = x \log p + (n-x) \log(1-p)$$
$$l_n(\hat{p}) = x \log \hat{p} + (n-x) \log(1-\hat{p})$$

Example: Profile CI

Let $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$ and $X = \sum_{i=1}^{n} X_i$. Then $X \sim \text{Binom}(n, p)$

We got from previous slide:

$$l_n(p) = x \log p + (n - x) \log(1 - p)$$
$$l_n(\hat{p}) = x \log \hat{p} + (n - x) \log(1 - \hat{p})$$

Now obtain the log-likelihood ratio

$$2\left[l_n\left(\hat{\theta}\right) - l_n(\theta)\right] = 2\left[x \log \frac{\hat{p}}{p} + (n - x) \log \frac{1 - \hat{p}}{1 - p}\right]$$

Choose the following set as CI:

$$S = \left\{\theta \in \Theta\colon\ 2l_n\left(\hat{\theta}\right) - 2l_n(\theta) < \chi_\alpha^2\right\}$$

Example: Profile CI

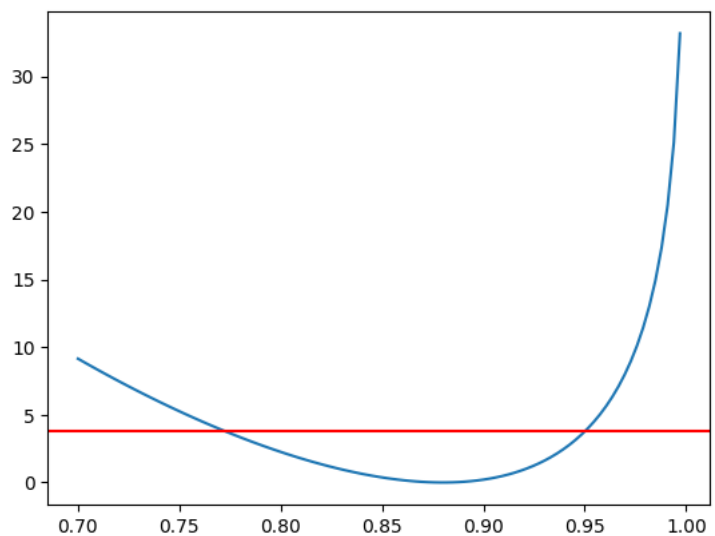
Let $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$ and $X = \sum_{i=1}^{n} X_i$. Then

$$X \sim \text{Binom}(n, p)$$

$$2\left[l_n(\hat{\theta}) - l_n(\theta)\right] = 2\left[x \log\frac{\hat{p}}{p} + (n - x)\log\frac{1-\hat{p}}{1-p}\right]$$

Choose the following set as CI:

$$S = \left\{\theta \in \Theta\colon\ 2l_n(\hat{\theta}) - 2l_n(\theta) < \chi_\alpha^2\right\}$$



Transformations

If (L, U) is a CI for θ and h is a monotone *increasing* function, then $(h(L), h(U))$ is a CI for $h(\theta)$.

Example:

If $X_1, \ldots, X_n \sim N(\mu, \sigma)$, then $\left(e^{\bar{x}-1.96\frac{\sigma}{\sqrt{n}}}, e^{\bar{x}+1.96\frac{\sigma}{\sqrt{n}}}\right)$ is a 95% CI for e^{μ}.

Note: If $X \sim N(\mu, \sigma)$ then $e^X \sim$ Lognormal (μ, σ) with median e^{μ}.